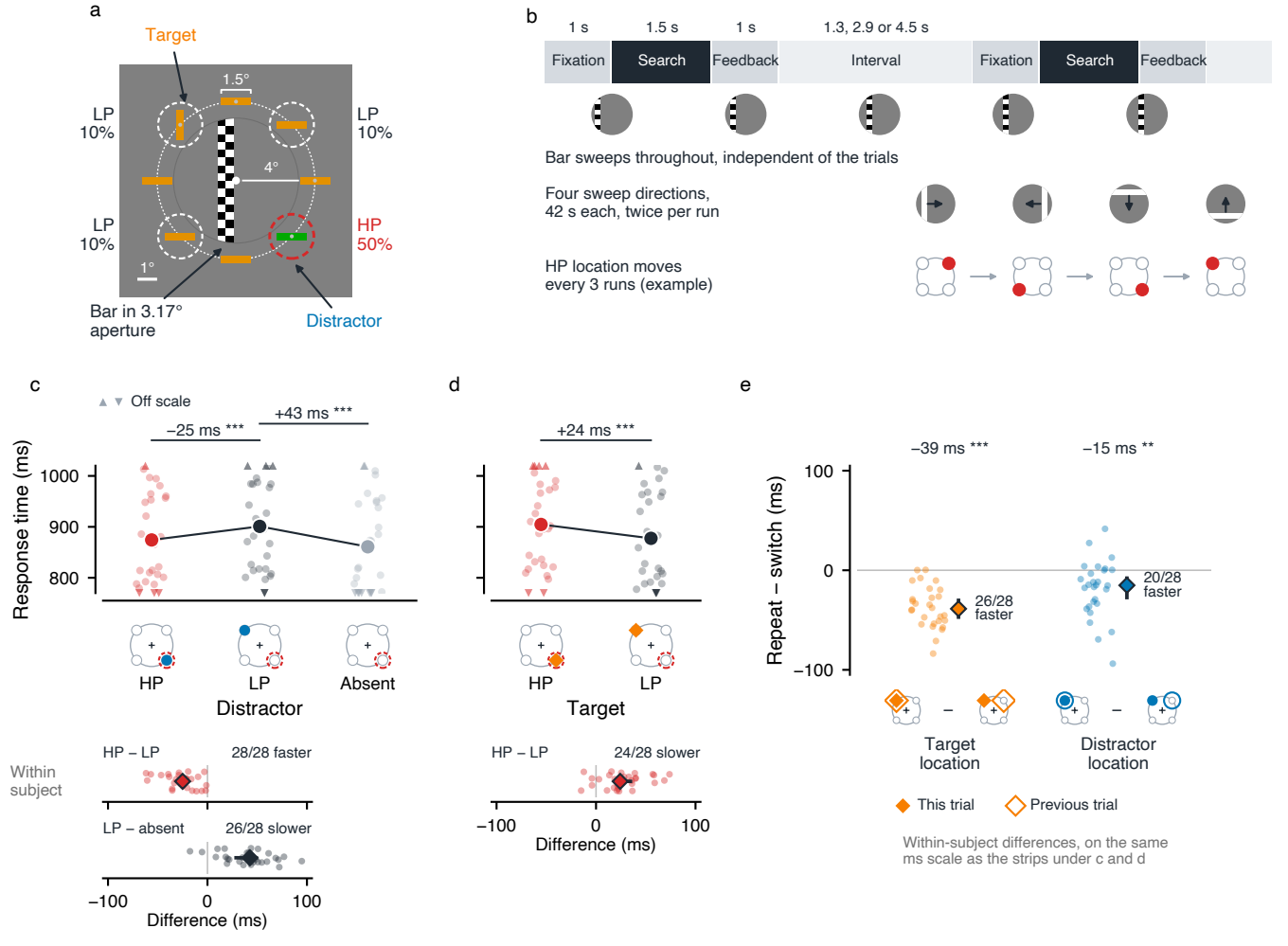
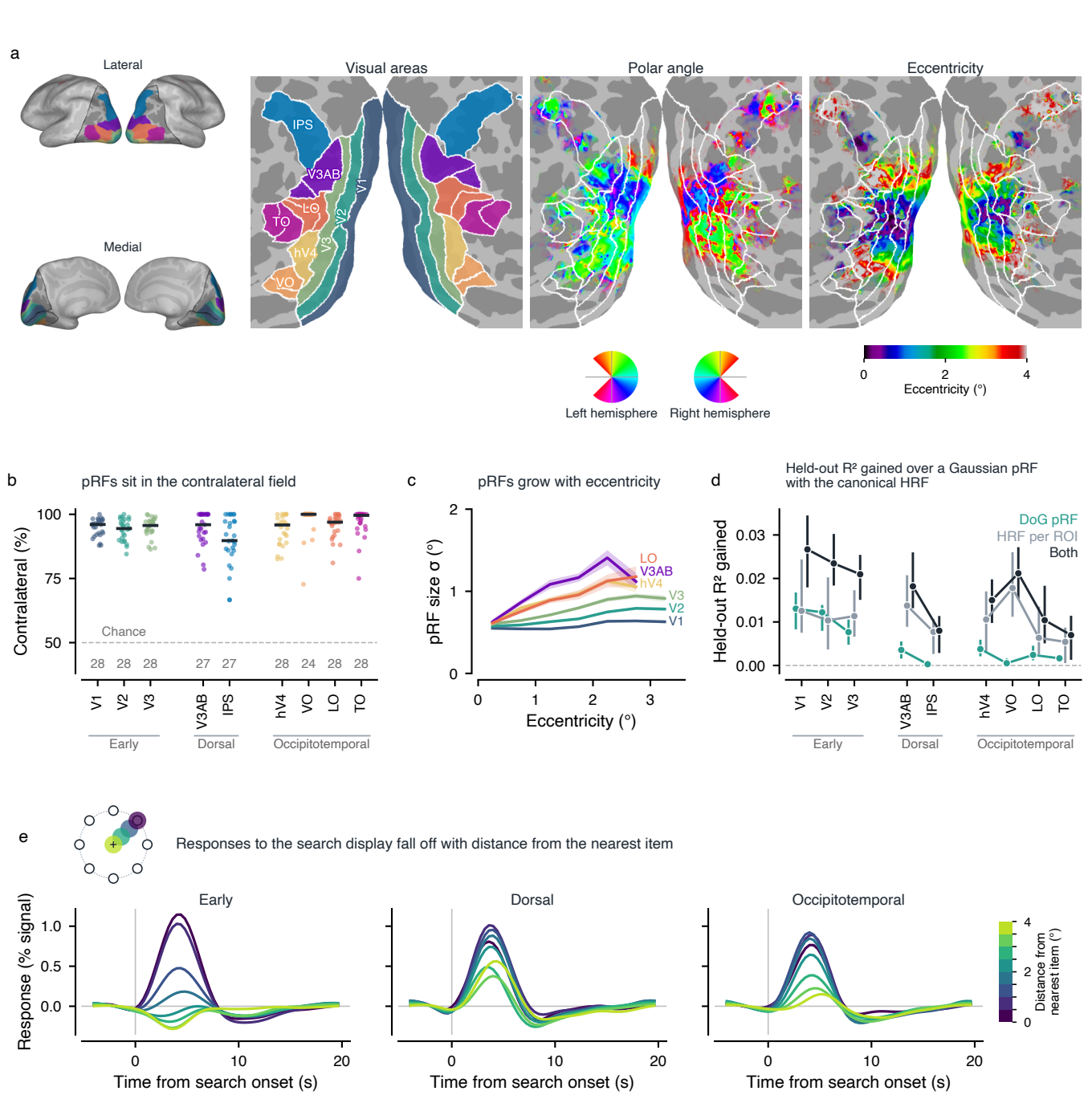
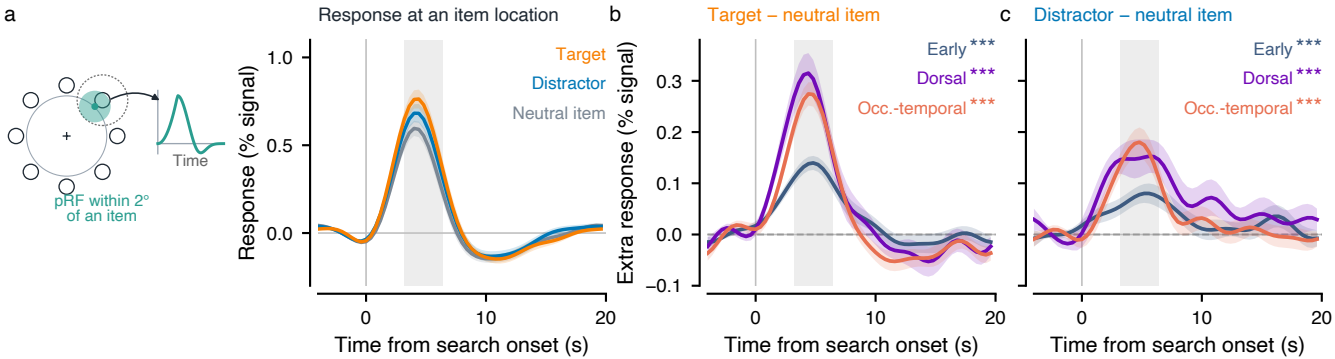
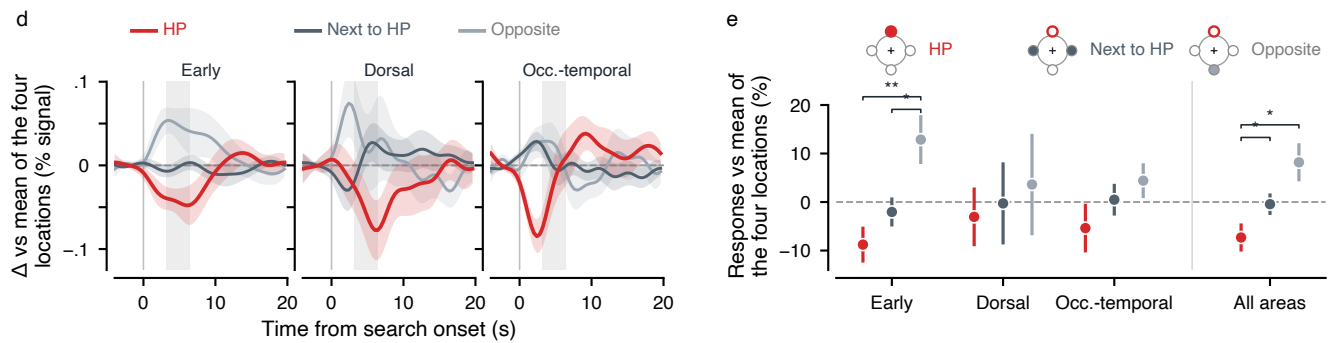

a
Target
1.5°
LP
10%
LP
10%
4°
LP
10%
HP
50%
1°
Bar in 3.17°
aperture
Distractor
b
1 s
1.5 s
1 s
1.3, 2.9 or 4.5 s
Fixation
Search
Feedback
Interval
Fixation
Search
Feedback
Bar sweeps throughout, independent of the trials
Four sweep directions,
42 s each, twice per run
HP location moves
every 3 runs (example)
c
Off scale
–25 ms ***
+43 ms ***
Response time (ms)
1000
900
800
HP
LP
Absent
Distractor
Within
subject
HP – LP
28/28 faster
LP – absent
26/28 slower
–100
0
100
Difference (ms)
d
+24 ms ***
HP
LP
Target
HP – LP
24/28 slower
–100
0
100
Difference (ms)
e
–39 ms ***
–15 ms **
Repeat – switch (ms)
100
0
–100
26/28
faster
20/28
faster
Target
location
Distractor
location
This trial
Previous trial
Within-subject differences, on the same
ms scale as the strips under c and d

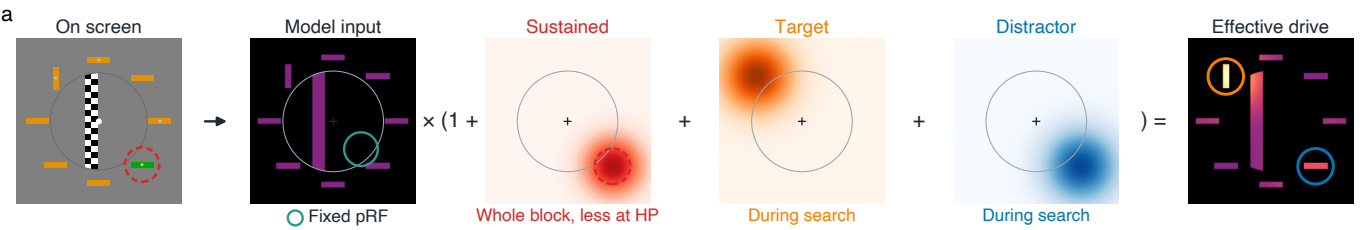

a
Lateral
Medial
Visual areas
Polar angle
Eccentricity
IPS
V3AB
LO
TO
hV4
VO
V1
V2
V3
Left hemisphere
Right hemisphere
0
2
4
Eccentricity (°)
b
pRFs sit in the contralateral field
Contralateral (%)
100
75
50
Chance
28 28 28 27 27 28 24 28 28
V1 V2 V3 V3AB IPS hV4 VO LO TO
Early
Dorsal
Occipitotemporal
c
pRFs grow with eccentricity
pRF size σ (°)
2
1
0
LO
V3AB
hV4
V3
V2
V1
0 1 2 3
Eccentricity (°)
d
Held-out R² gained over a Gaussian pRF with the canonical HRF
Held-out R² gained
0.03
0.02
0.01
0.00
DoG pRF
HRF per ROI
Both
V1 V2 V3 V3AB IPS hV4 VO LO TO
Early
Dorsal
Occipitotemporal
e
Responses to the search display fall off with distance from the nearest item
Early
Dorsal
Occipitotemporal
Response (% signal)
1.0
0.5
0.0
0 10 20
Time from search onset (s)
4
2
0
Distance from nearest item (°)

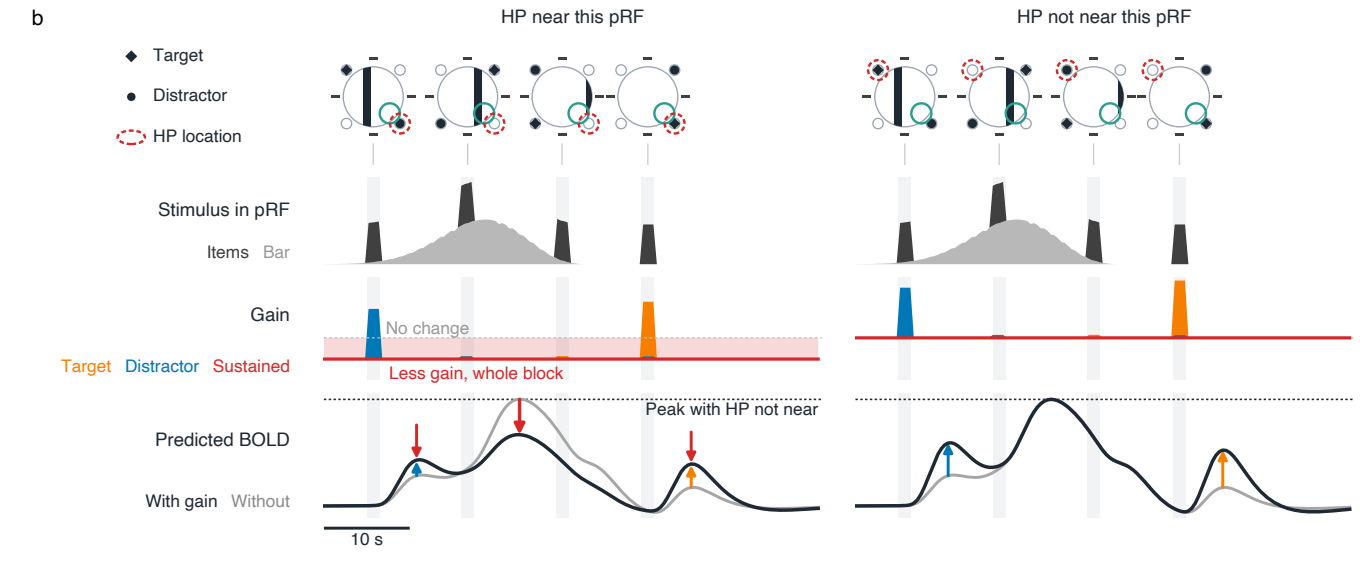
What the item is: more response to the target and the distractor than to a neutral item

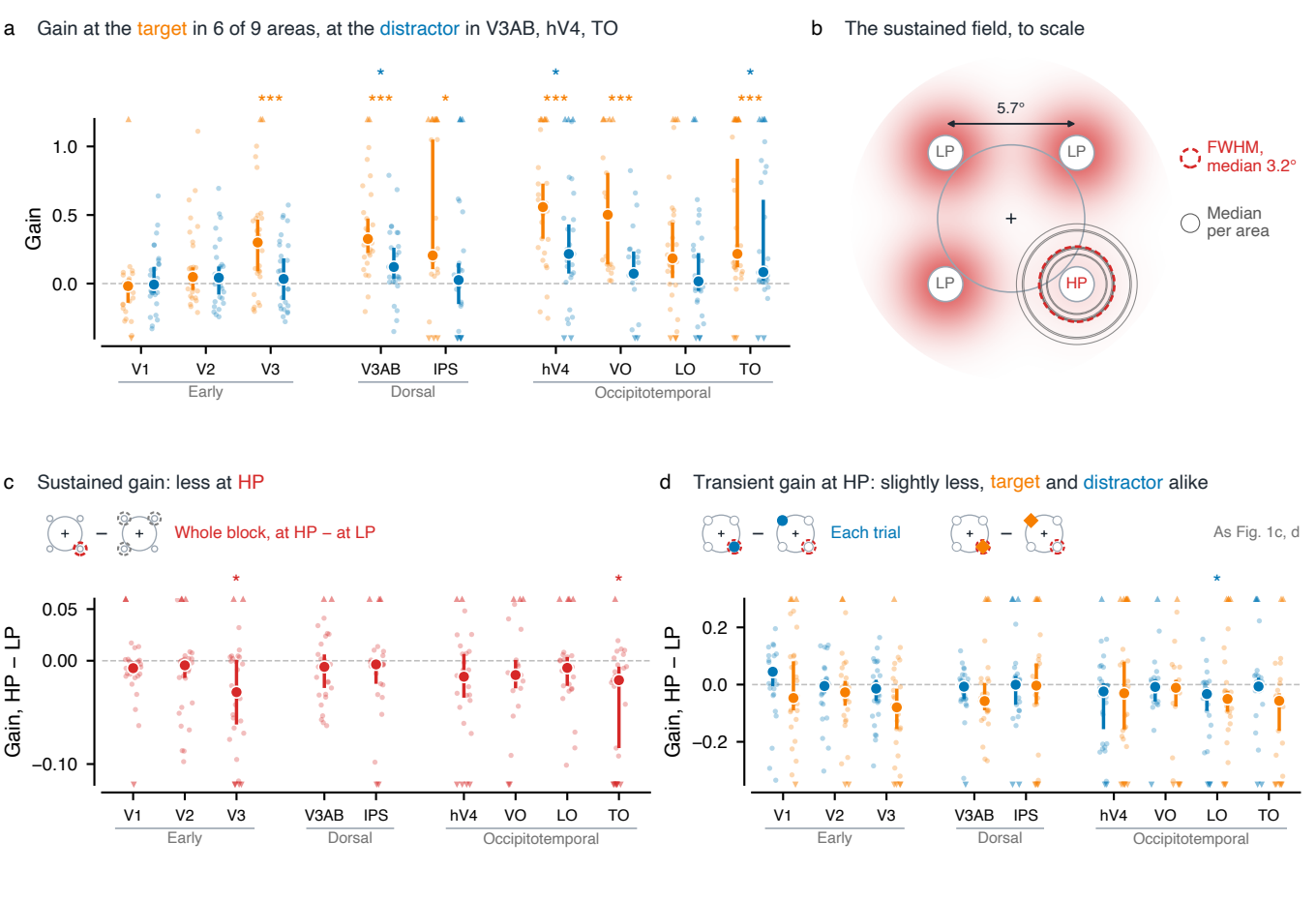


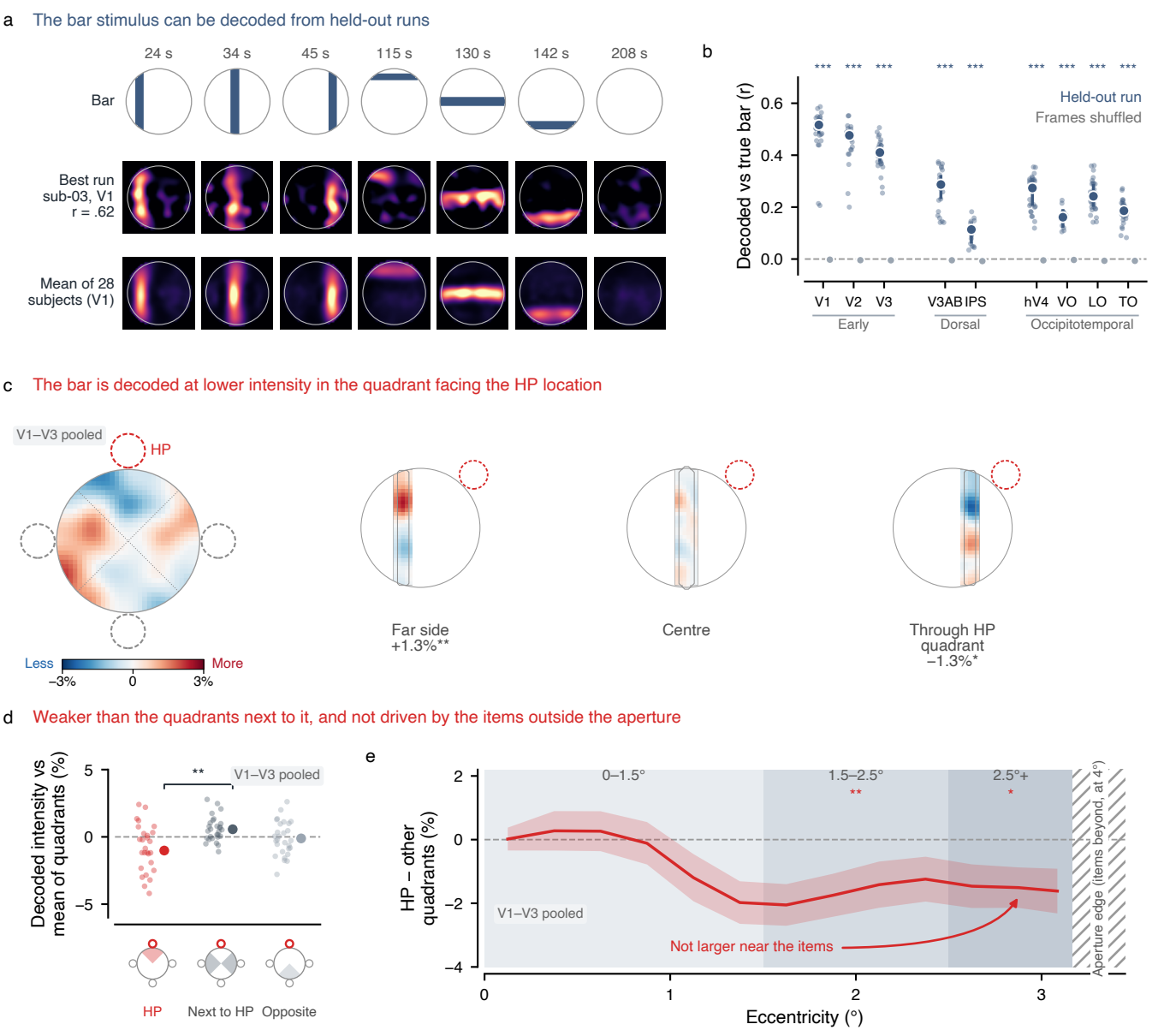
Where the item is: in early areas, least response at HP, less next to it than opposite it, for every item type

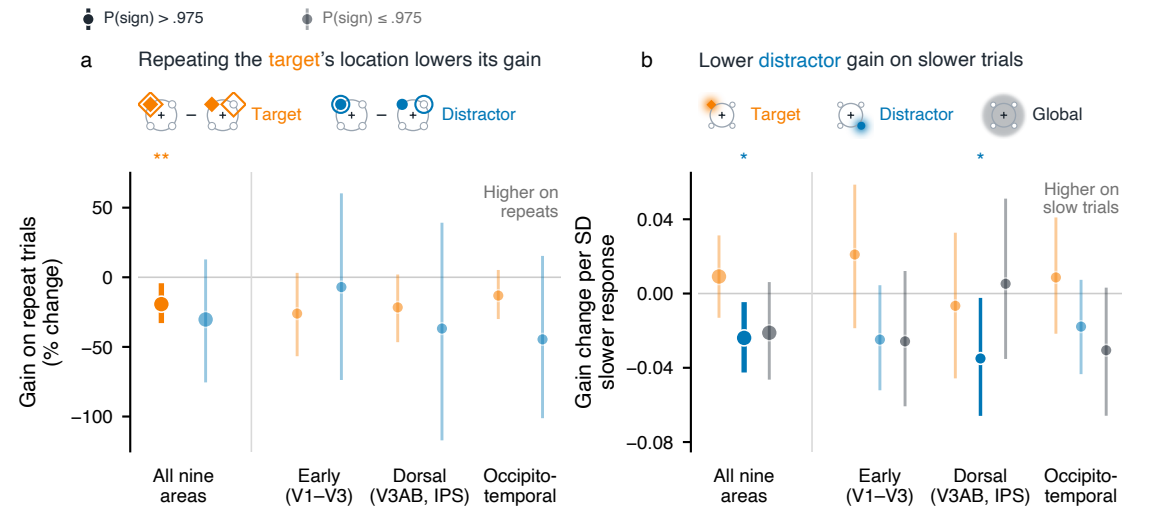



a
On screen
Model input
Sustained
Target
Distractor
Effective drive
× (1 +
+
+
) =
Fixed pRF
Whole block, less at HP
During search
During search


b
HP near this pRF
HP not near this pRF
Target
Distractor
HP location
Stimulus in pRF
Items Bar
Gain
No change
Target Distractor Sustained
Less gain, whole block
Predicted BOLD
Peak with HP not near
With gain Without
10 s

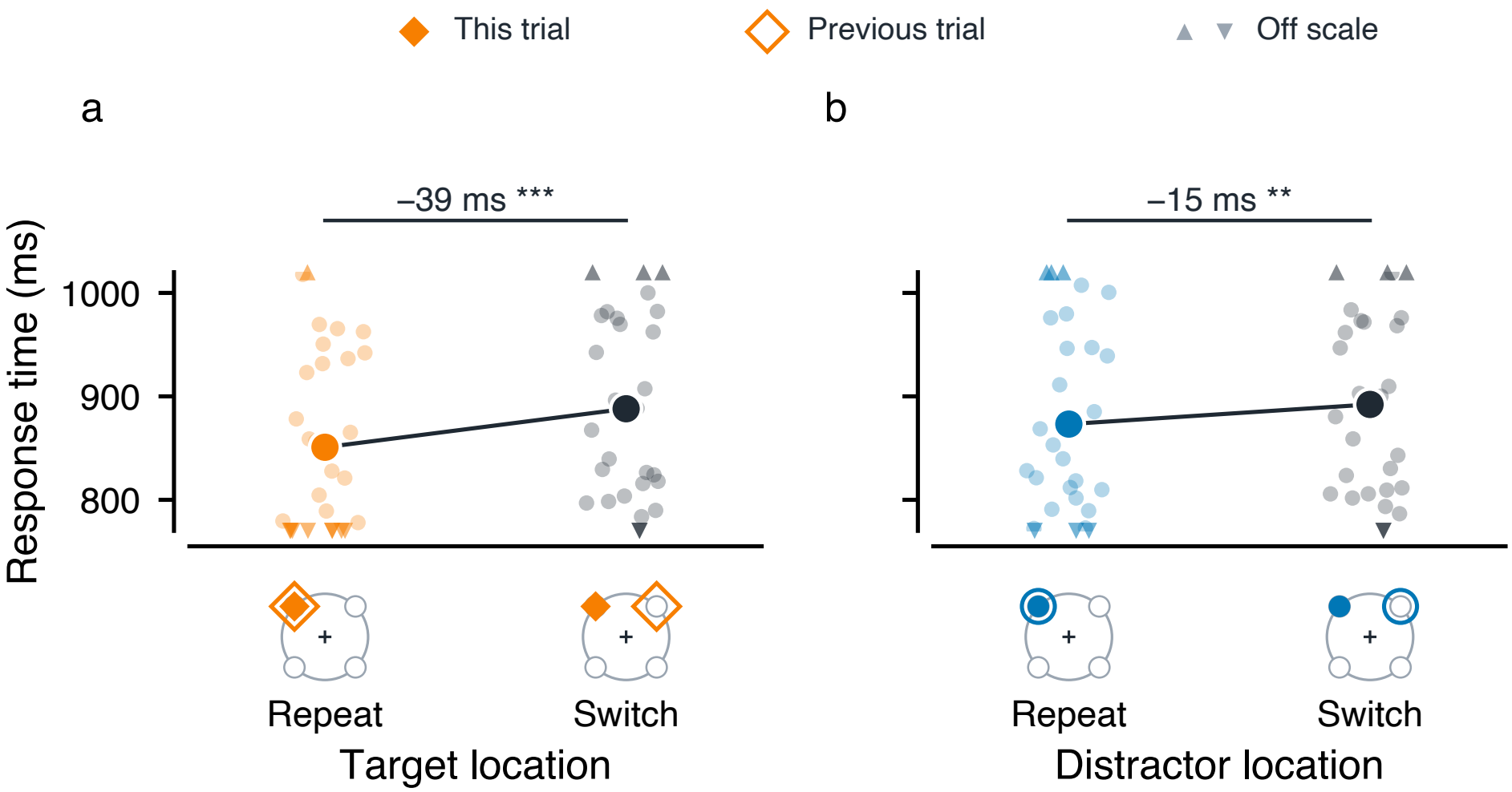

a
Gain at the target in 6 of 9 areas, at the distractor in V3AB, hV4, TO
Gain
1.0
0.5
0.0
V1
V2
V3
V3AB
IPS
hV4
VO
LO
TO
Early
Dorsal
Occipitotemporal
b
The sustained field, to scale
5.7°
LP
LP
LP
HP
+
FWHM, median 3.2°
Median per area
c
Sustained gain: less at HP
Whole block, at HP – at LP
Gain, HP – LP
0.05
0.00
−0.10
d
Transient gain at HP: slightly less, target and distractor alike
Each trial
As Fig. 1c, d
0.2
0.0
−0.2

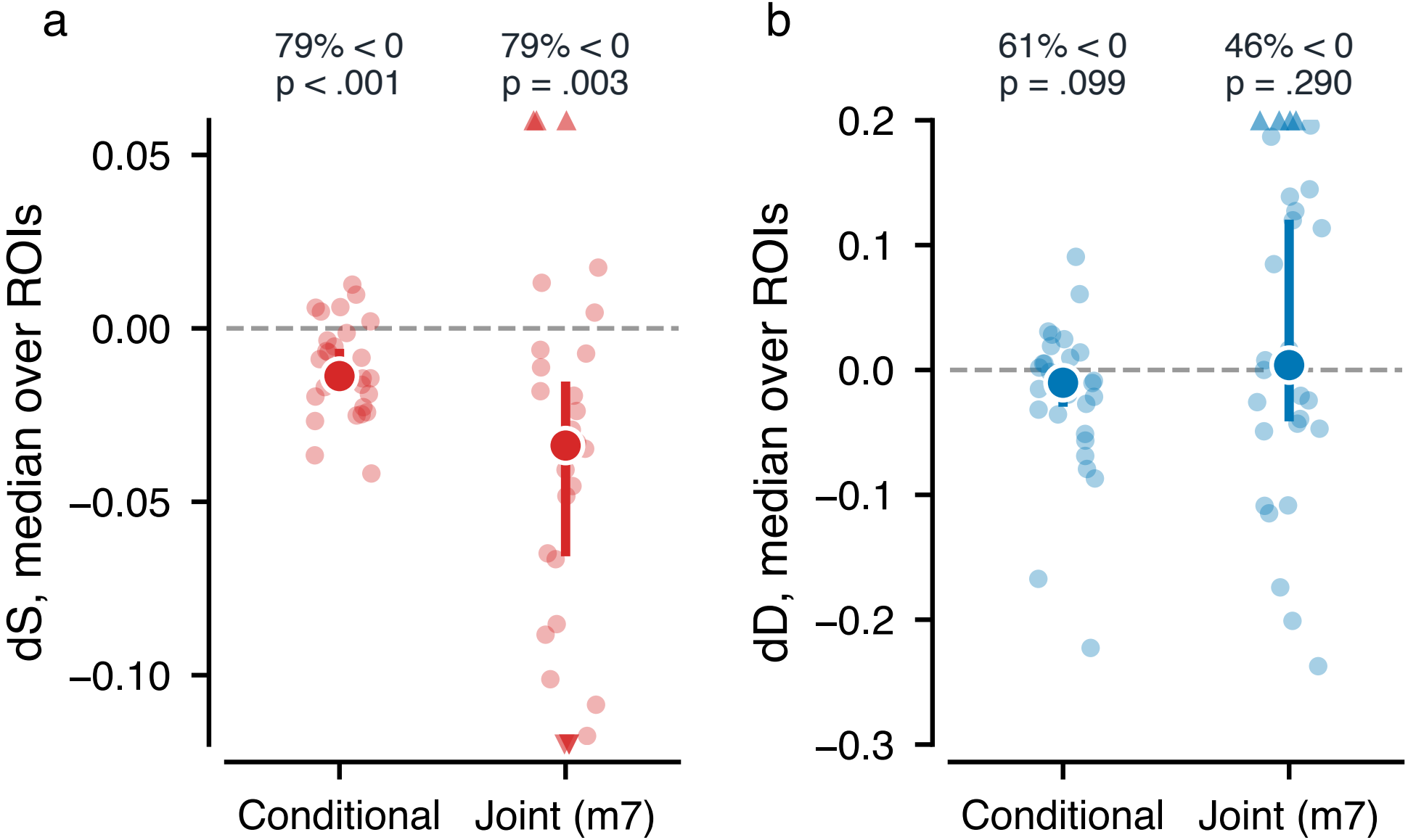

a
The bar stimulus can be decoded from held-out runs
24 s
34 s
45 s
115 s
130 s
142 s
208 s
Bar
Best run
sub-03, V1
r = .62
Mean of 28
subjects (V1)
b
Decoded vs true bar (r)
0.6
0.4
0.2
0.0
Held-out run
Frames shuffled
V1
V2
V3
V3AB
IPS
hV4
VO
LO
TO
Early
Dorsal
Occipitotemporal
c
The bar is decoded at lower intensity in the quadrant facing the HP location
V1–V3 pooled
HP
Less
More
−3%
0
3%
Far side
+1.3%**
Centre
Through HP
quadrant
−1.3%*
d
Weaker than the quadrants next to it, and not driven by the items outside the aperture
Decoded intensity vs
mean of quadrants (%)
5
0
−5
**
V1–V3 pooled
HP
Next to HP
Opposite
e
HP − other
quadrants (%)
2
0
−2
−4
0–1.5°
1.5–2.5°
**
2.5°+
*
V1–V3 pooled
Not larger near the items
Aperture edge (items beyond, at 4°)
0
1
2
3
Eccentricity (°)

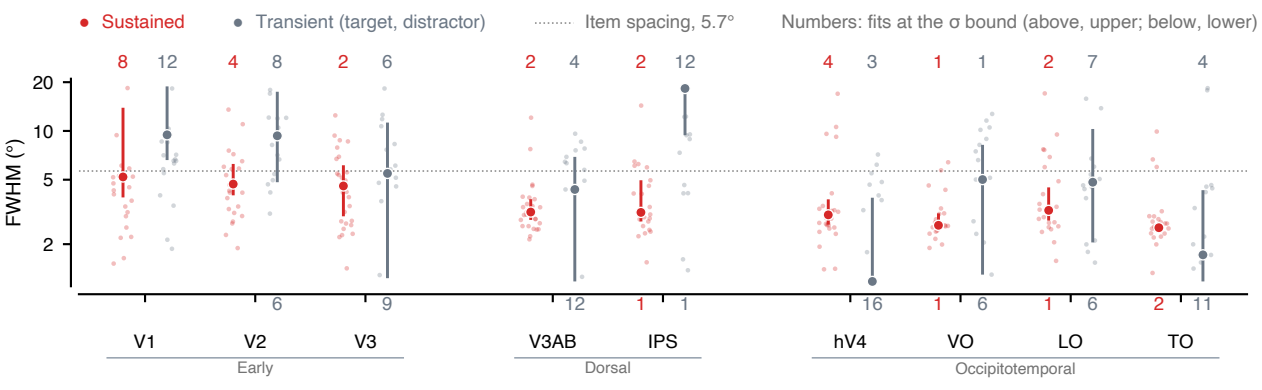

P(sign) > .975
P(sign) ≤ .975
a
Repeating the target's location lowers its gain
Target
Distractor
**
Higher on repeats
Gain on repeat trials (% change)
50
0
−50
−100
All nine areas
Early (V1–V3)
Dorsal (V3AB, IPS)
Occipito-temporal
b
Lower distractor gain on slower trials
Target
Distractor
Global
*
*
Higher on slow trials
Gain change per SD slower response
0.04
0.00
−0.04
−0.08
All nine areas
Early (V1–V3)
Dorsal (V3AB, IPS)
Occipito-temporal

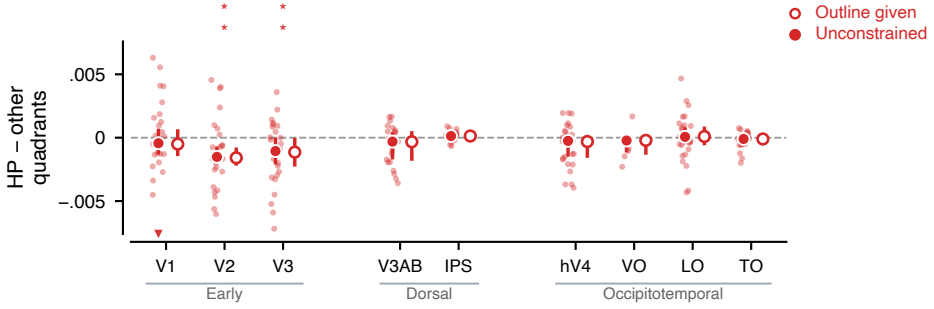
Model	Parent	Adds	Contrast	DoG: better; direction	Gaussian: better; direction	DN: better; direction
m00		pRF on the bar				
m0	m00	Search items, weight fitted	Items	100% (p = 7.5e-09)	93% (p = 3.7e-08); 89% (p = 1.4e-07)	100% (p = 7.5e-09)
m1	m0	Target gain	T	75% (p = 0.00072); 96% (p = 2.2e-08)	89% (p = 3.7e-06); 93% (p = 3.3e-05)	100% (p = 7.5e-09)
m2	m0	Distractor gain, same at all locations	D	61% (p = 0.16); 75% (p = 0.034)	79% (p = 0.03); 86% (p = 0.018)	32% (p = 0.12)
m3	m0	Sustained gain, same at all locations	S	93% (p = 1.5e-06); 100% (p = 7.5e-09)	89% (p = 1.1e-07); 100% (p = 7.5e-09)	93% (p = 1e-06)
m3b	m1	Distractor gain	D given T	61% (p = 0.33); 75% (p = 0.047)	64% (p = 0.88); 68% (p = 0.15)	82% (p = 0.0082)
m4	m3b	Sustained gain				
m5	m4	Sustained gain differs at HP	dS	68% (p = 0.41); 75% (p = 0.071)	71% (p = 0.54); 79% (p = 0.044)	71% (p = 0.038)
m6	m4	Distractor gain differs at HP	dD	43% (p = 0.25); 50% (p = 0.96)	50% (p = 0.87); 57% (p = 0.26)	71% (p = 0.0052)
m7	m5	Both HP-specific terms				
m8	m4	Target gain differs at HP	dT	61% (p = 0.25); 71% (p = 0.014)	71% (p = 0.73); 68% (p = 0.22)	57% (p = 0.75)

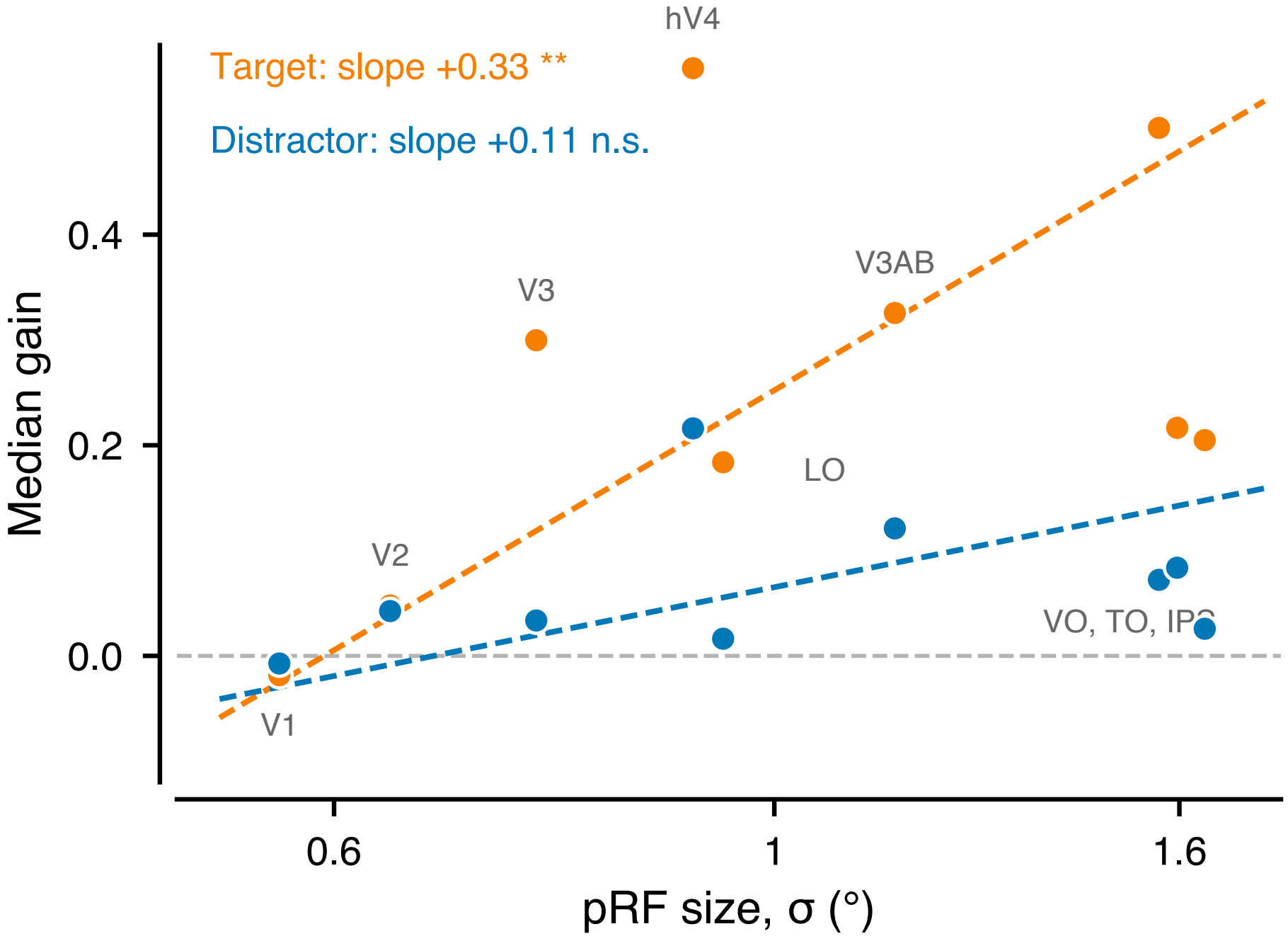
How much each term adds to out-of-subject prediction (leave-one-subject-out, final fits; each pRF model with its own jointly fitted HRF per subject × ROI). Better: share of 30 subjects whose ΔR² on held-out data is positive when the term is added at the median of the other 29 subjects' own estimates (signal voxels; per subject the median over ROIs). Wilcoxon signed-rank p. Direction: sign-flip test, ΔR² at +δ minus at −δ, which removes the cost of adding a term (a true 50% null). Items: the item


This trial
Previous trial
Off scale
a
b
−39 ms ***
−15 ms **
Response time (ms)
1000
900
800
Repeat
Switch
Target location
Repeat
Switch
Distractor location


a
79% < 0
p < .001
79% < 0
p = .003
0.05
0.00
−0.05
−0.10
dS, median over ROIs
Conditional
Joint (m7)
b
61% < 0
p = .099
46% < 0
p = .290
0.2
0.1
0.0
−0.1
−0.2
−0.3
dD, median over ROIs
Conditional
Joint (m7)

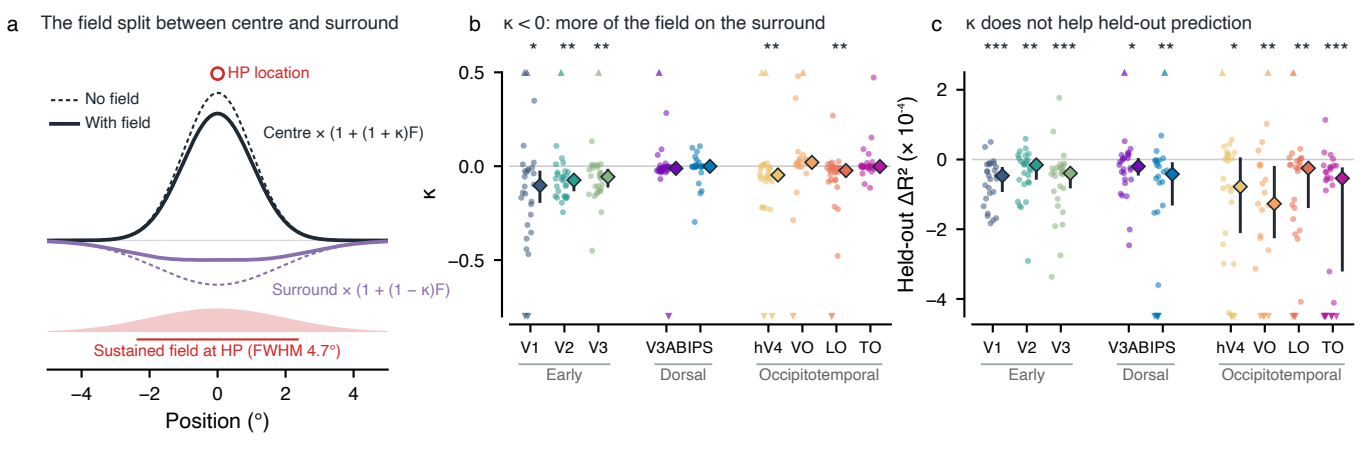

Sustained
Transient (target, distractor)
Item spacing, 5.7°
Numbers: fits at the σ bound (above, upper; below, lower)
FWHM (°)
20
10
5
2
8 12
4 8
2 6
2 4
2 12
4 3
1 1
2 7
4
6
9
12
1
1
16
1
6
1
6
2
11
V1
V2
V3
V3AB
IPS
hV4
VO
LO
TO
Early
Dorsal
Occipitotemporal

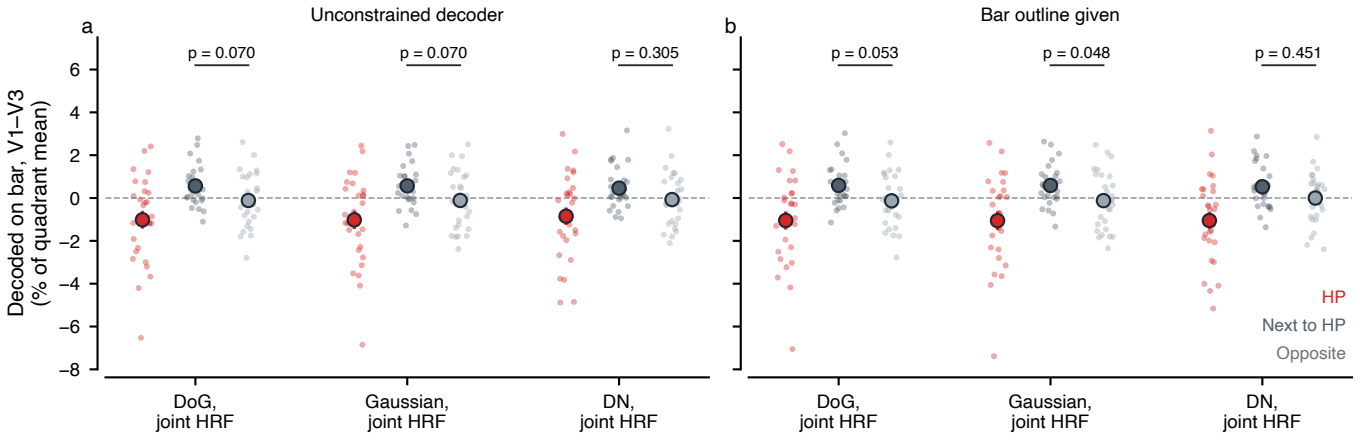

Outline given
Unconstrained
HP – other quadrants
.005
0
–.005
V1
V2
V3
V3AB
IPS
hV4
VO
LO
TO
Early
Dorsal
Occipitotemporal

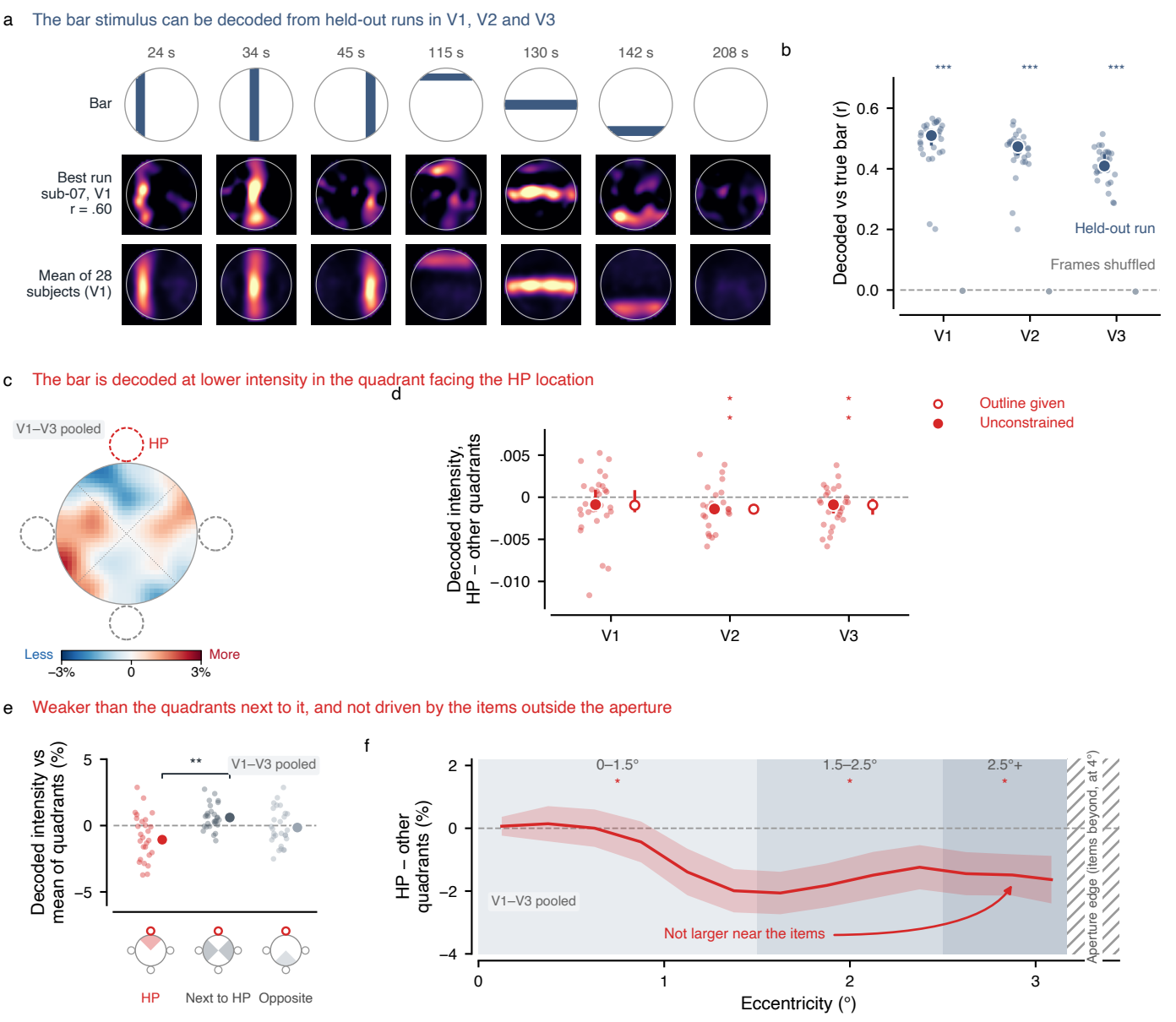

Target: slope +0.33 **
Distractor: slope +0.11 n.s.
hV4
V3
V3AB
LO
V2
V1
VO, TO, IP6
Median gain
0.4
0.2
0.0
0.6
1
1.6
pRF size, σ (°)

Predicts new subjects (% of subjects)

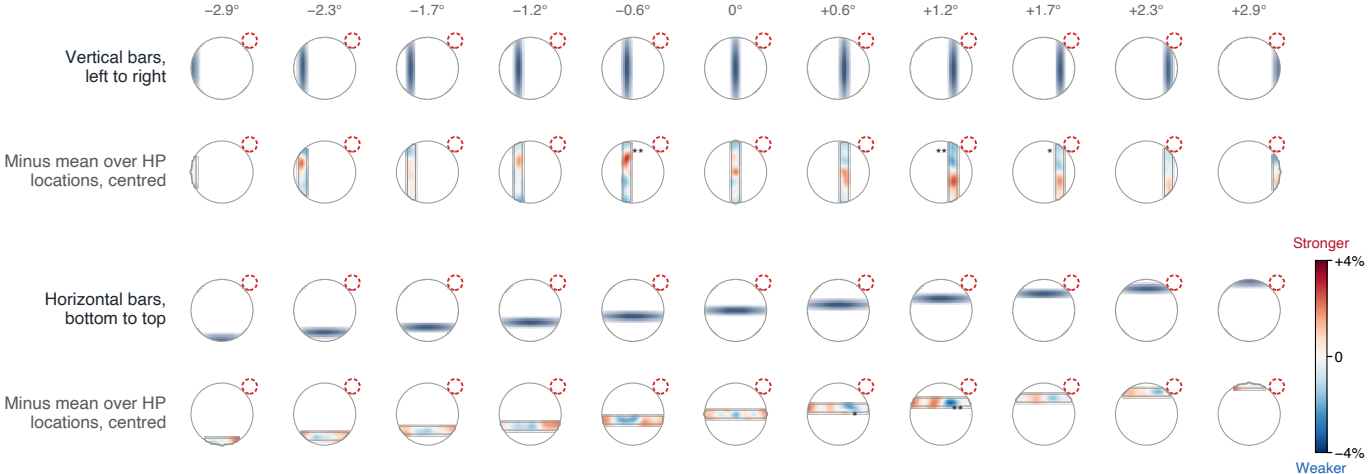
	Target	Distractor (with target)	HP – LP Sustained	HP – LP Distractor
V1	57	46	75	61
V2	57	54	57	4
V3	96	54	68	57
V3AB	100	83	70	48
IPS	92	62	69	0
hV4	92	84	76	68
VO	100	71	71	57
LO	78	0	56	70
TO	83	83	75	54

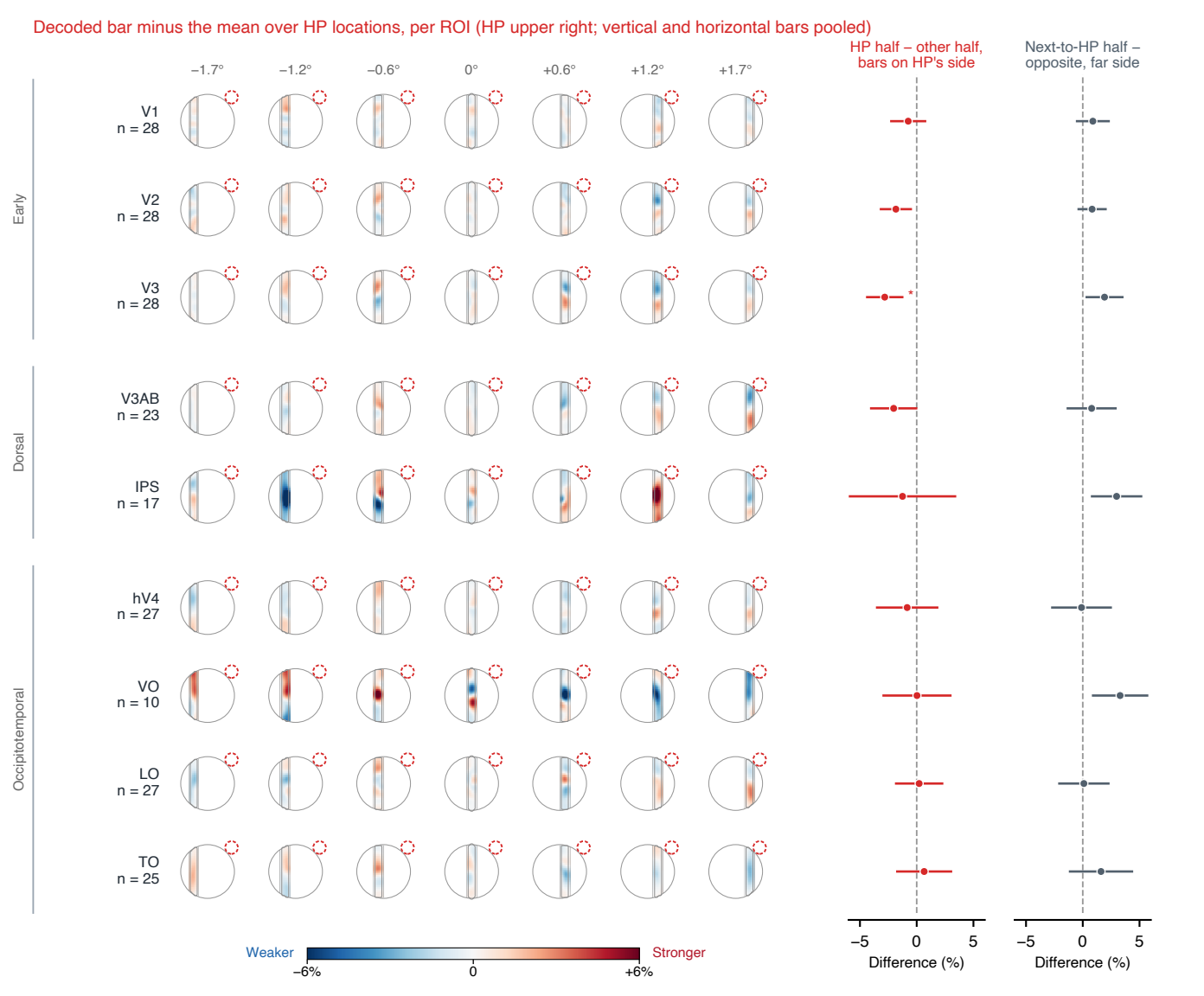

a
The field split between centre and surround
HP location
No field
With field
Centre × (1 + (1 + κ)F)
Surround × (1 + (1 − κ)F)
Sustained field at HP (FWHM 4.7°)
−4
−2
0
2
4
Position (°)
b
κ < 0: more of the field on the surround
κ
0.5
0.0
−0.5
*
**
**
**
**
V1
V2
V3
V3ABIPS
hV4
VO
LO
TO
Early
Dorsal
Occipitotemporal
c
κ does not help held-out prediction
Held-out ΔR² (× 10⁻⁴)
2
0
−2
−4
***
**
***
*
**
*
**
**
***
V1
V2
V3
V3ABIPS
hV4
VO
LO
TO
Early
Dorsal
Occipitotemporal


a
Unconstrained decoder
b
Bar outline given
p = 0.070
p = 0.070
p = 0.305
p = 0.053
p = 0.048
p = 0.451
Decoded on bar, V1–V3
(% of quadrant mean)
6
4
2
0
−2
−4
−6
−8
HP
Next to HP
Opposite
DoG,
joint HRF
Gaussian,
joint HRF
DN,
joint HRF
DoG,
joint HRF
Gaussian,
joint HRF
DN,
joint HRF

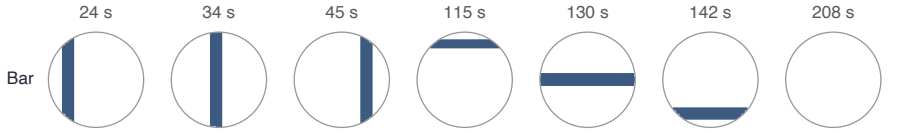

a
The bar stimulus can be decoded from held-out runs in V1, V2 and V3
24 s
34 s
45 s
115 s
130 s
142 s
208 s
Bar
Best run
sub-07, V1
r = .60
Mean of 28
subjects (V1)
b
Decoded vs true bar (r)
0.6
0.4
0.2
0.0
***
***
***
Held-out run
Frames shuffled
V1
V2
V3
c
The bar is decoded at lower intensity in the quadrant facing the HP location
V1–V3 pooled
HP
Less
More
–3%
0
3%
d
Outline given
Unconstrained
Decoded intensity,
HP – other quadrants
.005
0
–.005
–.010
V1
V2
V3
e
Weaker than the quadrants next to it, and not driven by the items outside the aperture
Decoded intensity vs
mean of quadrants (%)
5
0
–5
**
V1–V3 pooled
HP
Next to HP
Opposite
f
HP – other
quadrants (%)
2
0
–2
–4
0–1.5°
1.5–2.5°
2.5°+
V1–V3 pooled
Not larger near the items
Aperture edge (items beyond, at 4°)
0
1
2
3
Eccentricity (°)

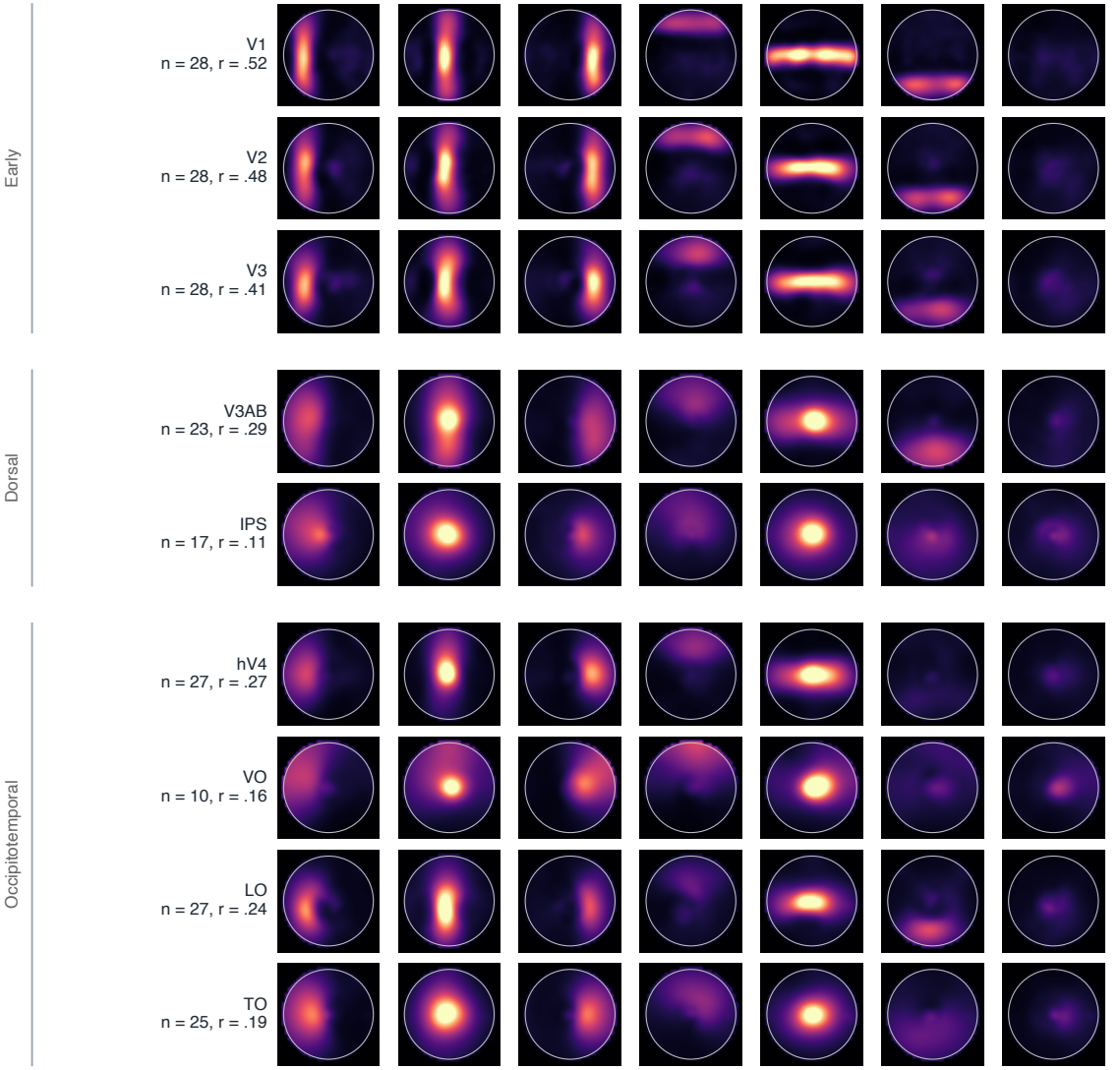
The same bar is decoded weaker where it passes the quadrant facing HP, for vertical and horizontal bars (V1–V3)

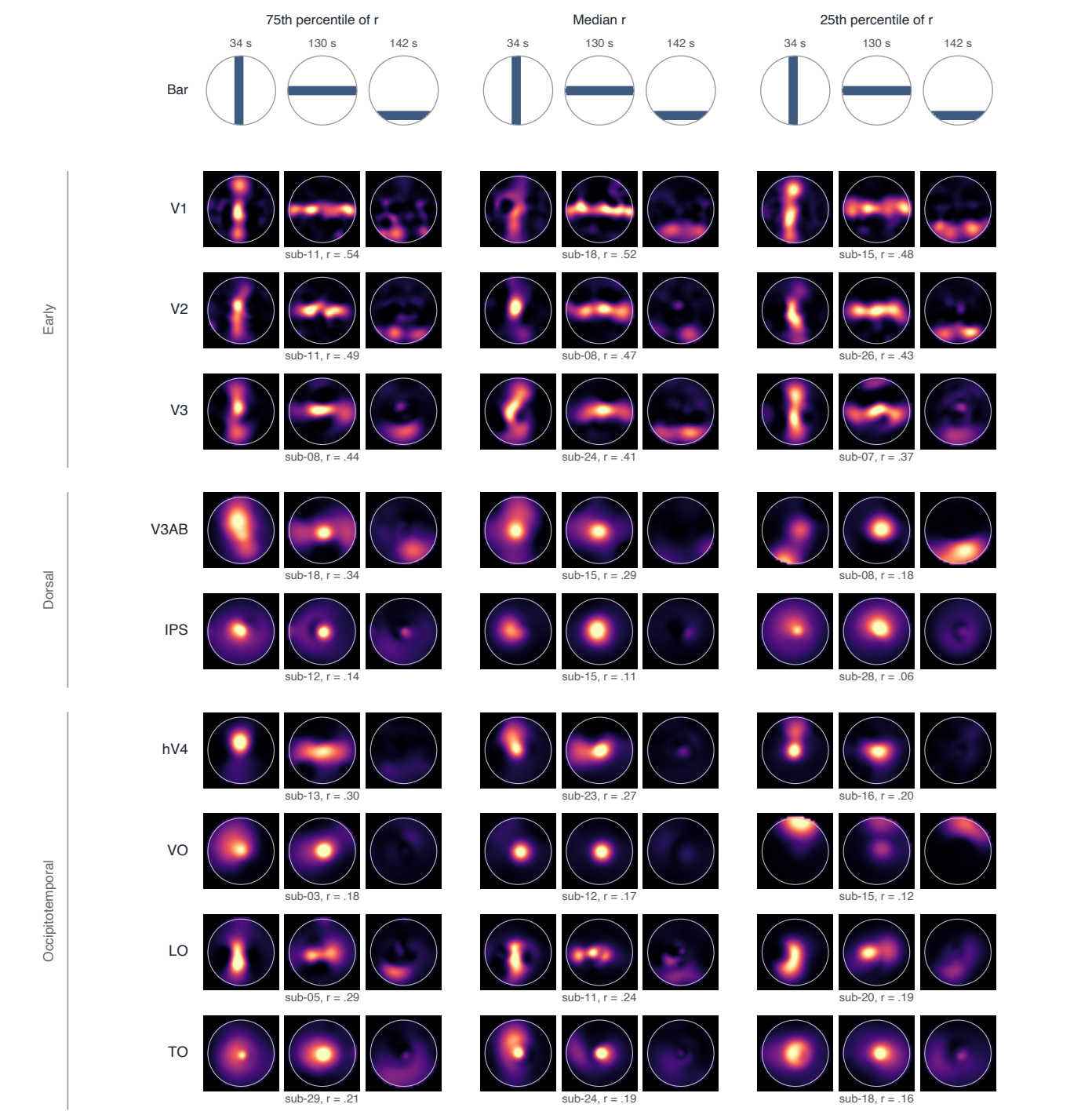


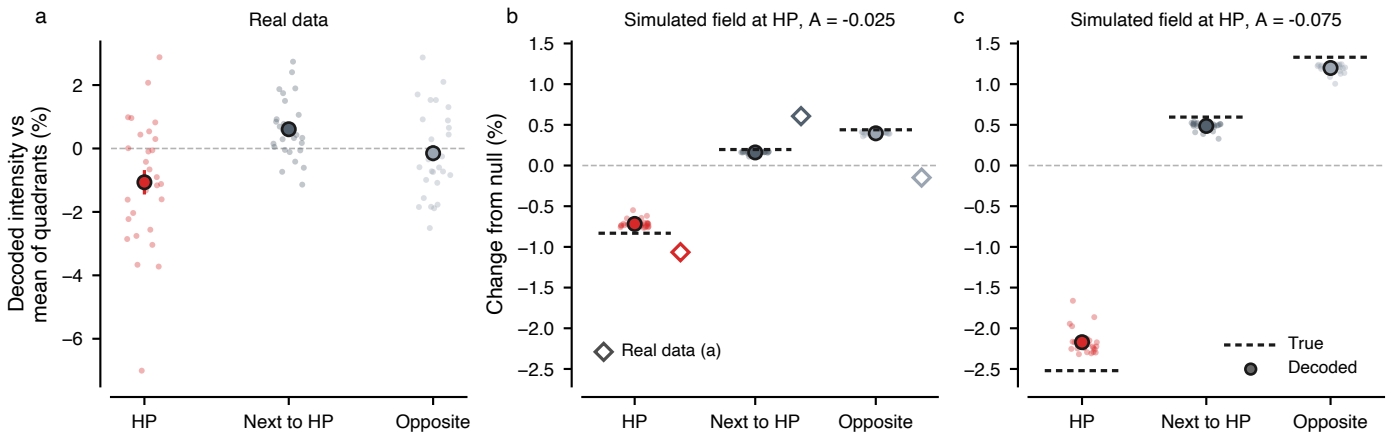

Decoded bar minus the mean over HP locations, per ROI (HP upper right; vertical and horizontal bars pooled)
−1.7°
−1.2°
−0.6°
0°
+0.6°
+1.2°
+1.7°
Early
V1
n = 28
V2
n = 28
V3
n = 28
Dorsal
V3AB
n = 23
IPS
n = 17
Occipitotemporal
hV4
n = 27
VO
n = 10
LO
n = 27
TO
n = 25
Weaker
Stronger
−6%
0
+6%
HP half − other half, bars on HP's side
Next-to-HP half − opposite, far side
*
−5
0
5
Difference (%)
−5
0
5
Difference (%)

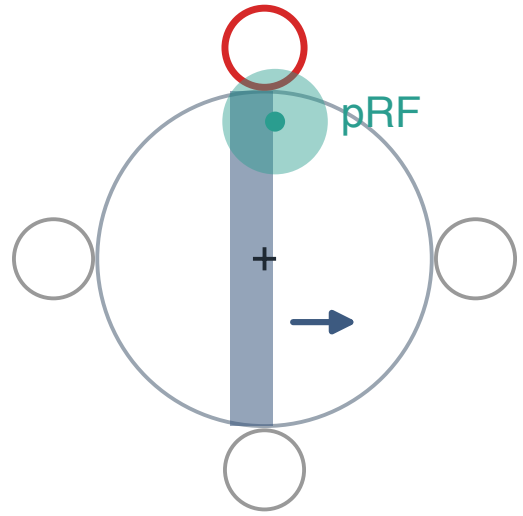
The bar decoded from each area: mean of the subjects' decoded frames

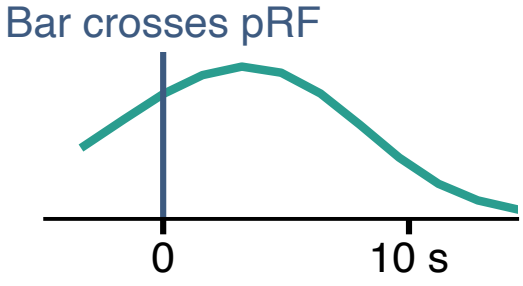


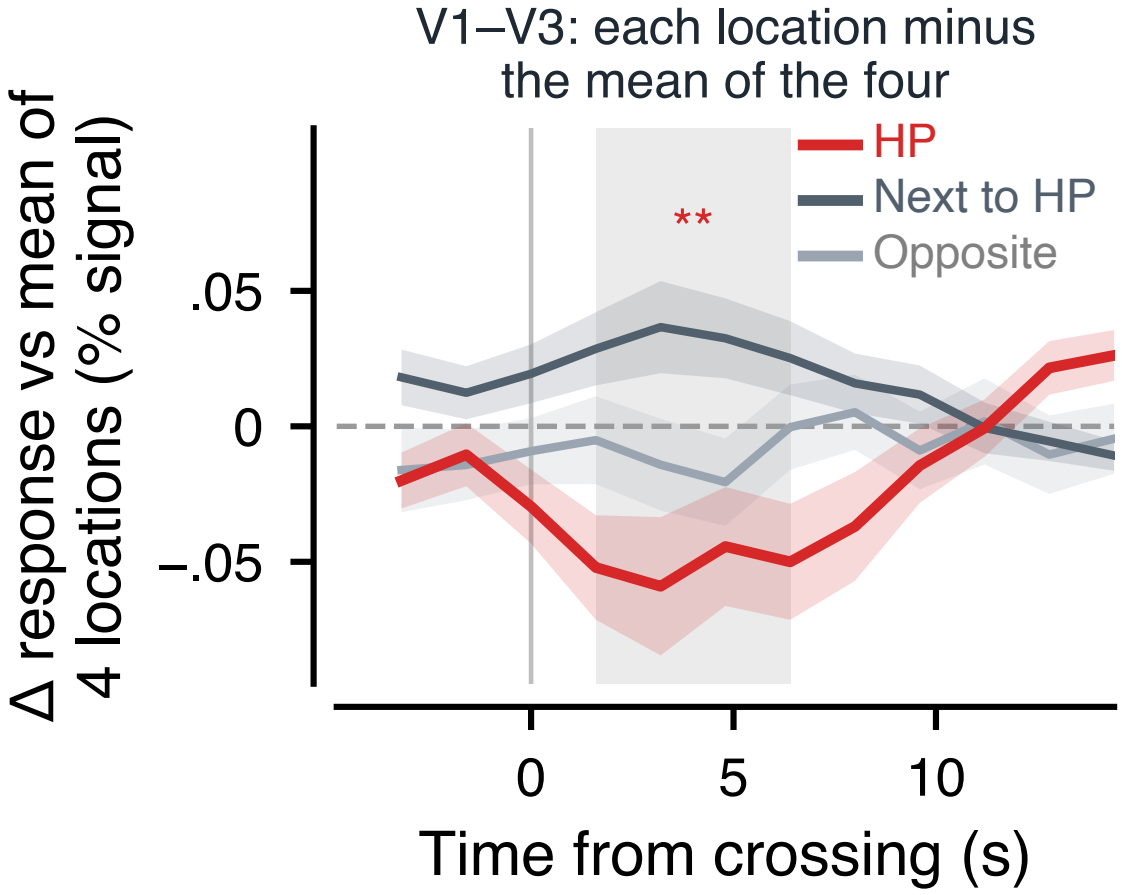


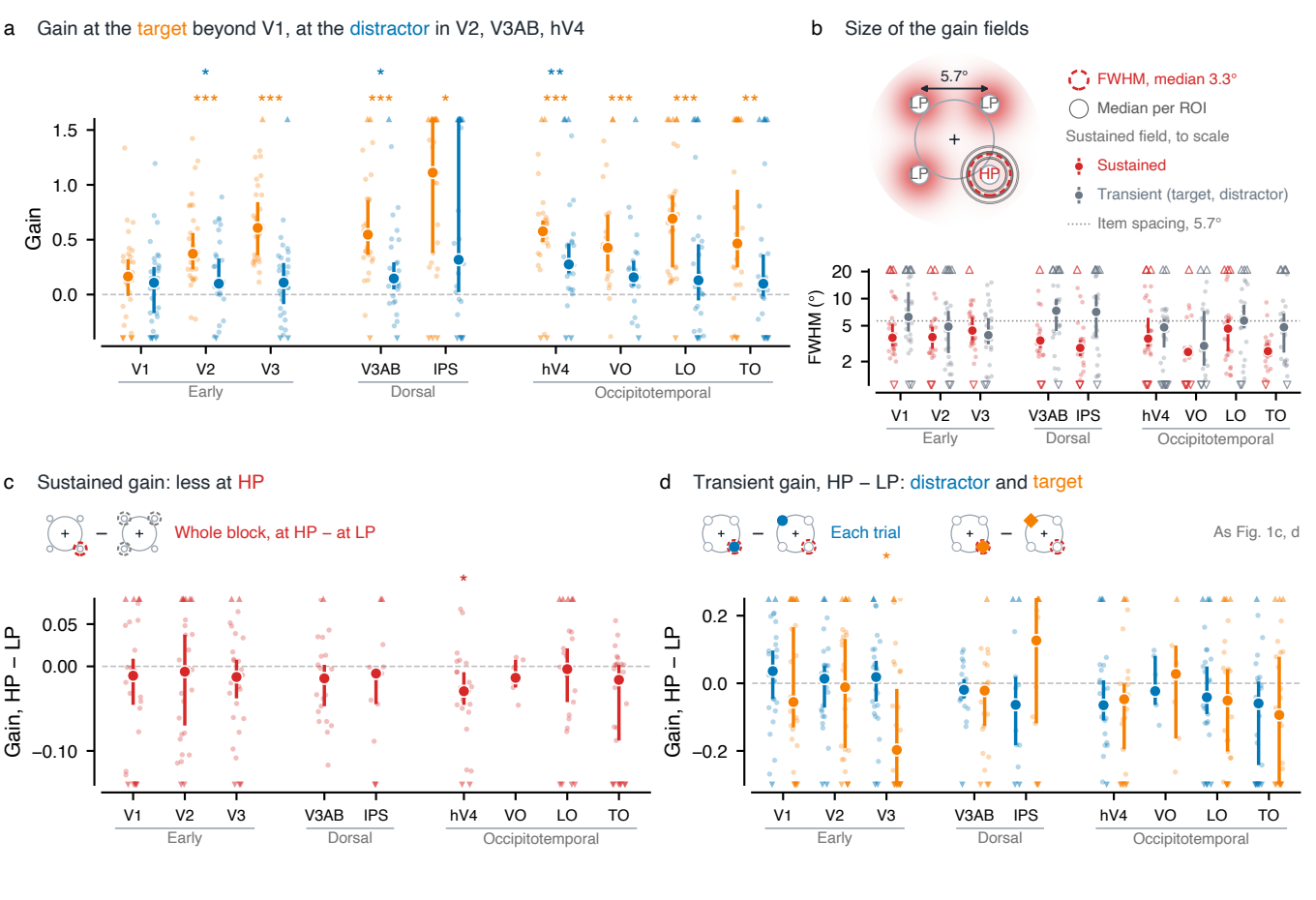

75th percentile of r
Median r
25th percentile of r
34 s
130 s
142 s
Bar
Early
V1
sub-11, r = .54
sub-18, r = .52
sub-15, r = .48
V2
sub-11, r = .49
sub-08, r = .47
sub-26, r = .43
V3
sub-08, r = .44
sub-24, r = .41
sub-07, r = .37
Dorsal
V3AB
sub-18, r = .34
sub-15, r = .29
sub-08, r = .18
IPS
sub-12, r = .14
sub-15, r = .11
sub-28, r = .06
Occipitotemporal
hV4
sub-13, r = .30
sub-23, r = .27
sub-16, r = .20
VO
sub-03, r = .18
sub-12, r = .17
sub-15, r = .12
LO
sub-05, r = .29
sub-11, r = .24
sub-20, r = .19
TO
sub-29, r = .21
sub-24, r = .19
sub-18, r = .16

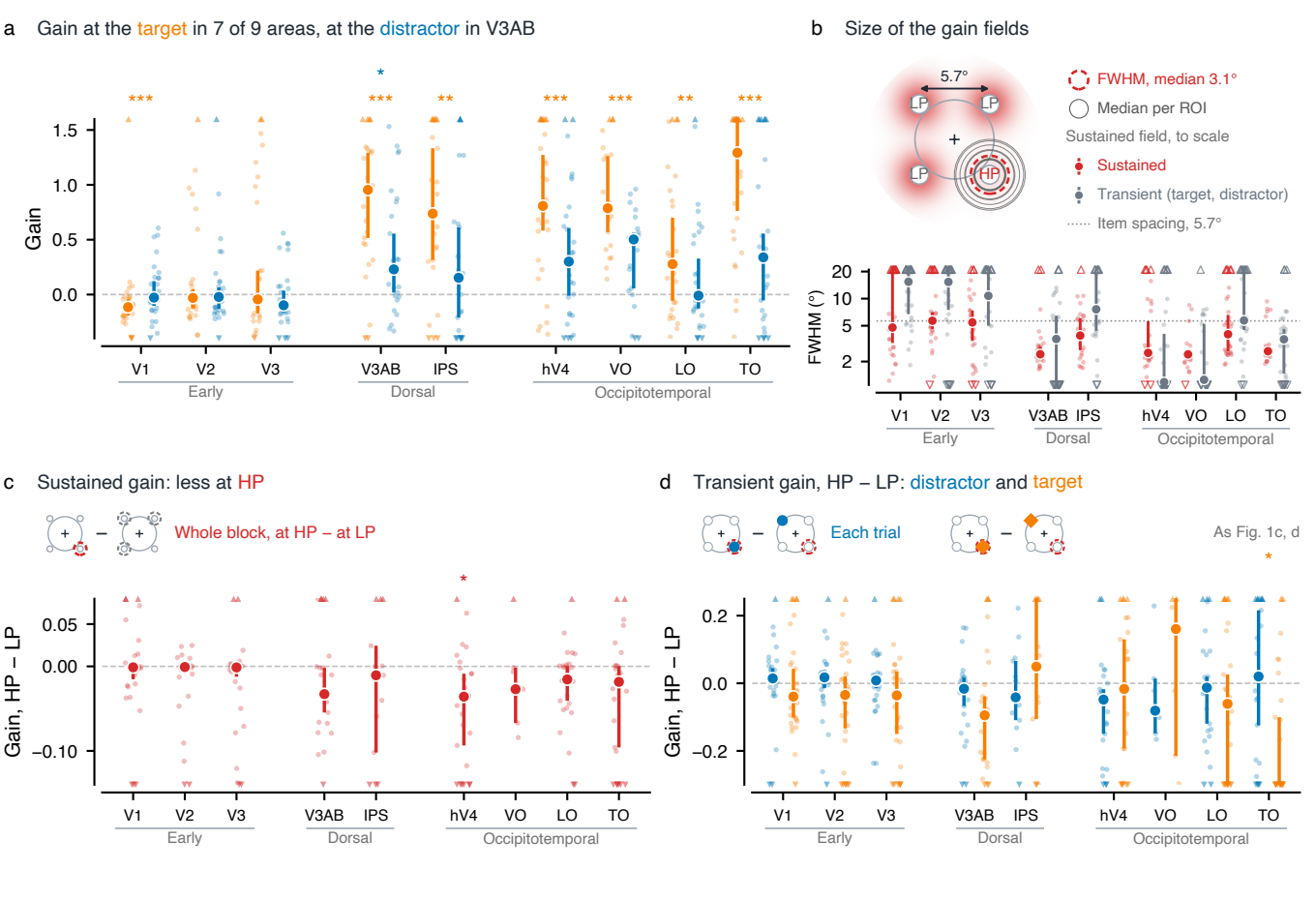

a
Real data
Decoded intensity vs mean of quadrants (%)
2
0
−2
−4
−6
HP
Next to HP
Opposite
b
Simulated field at HP, A = -0.025
Change from null (%)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
Real data (a)
HP
Next to HP
Opposite
c
Simulated field at HP, A = -0.075
True
Decoded
HP
Next to HP
Opposite

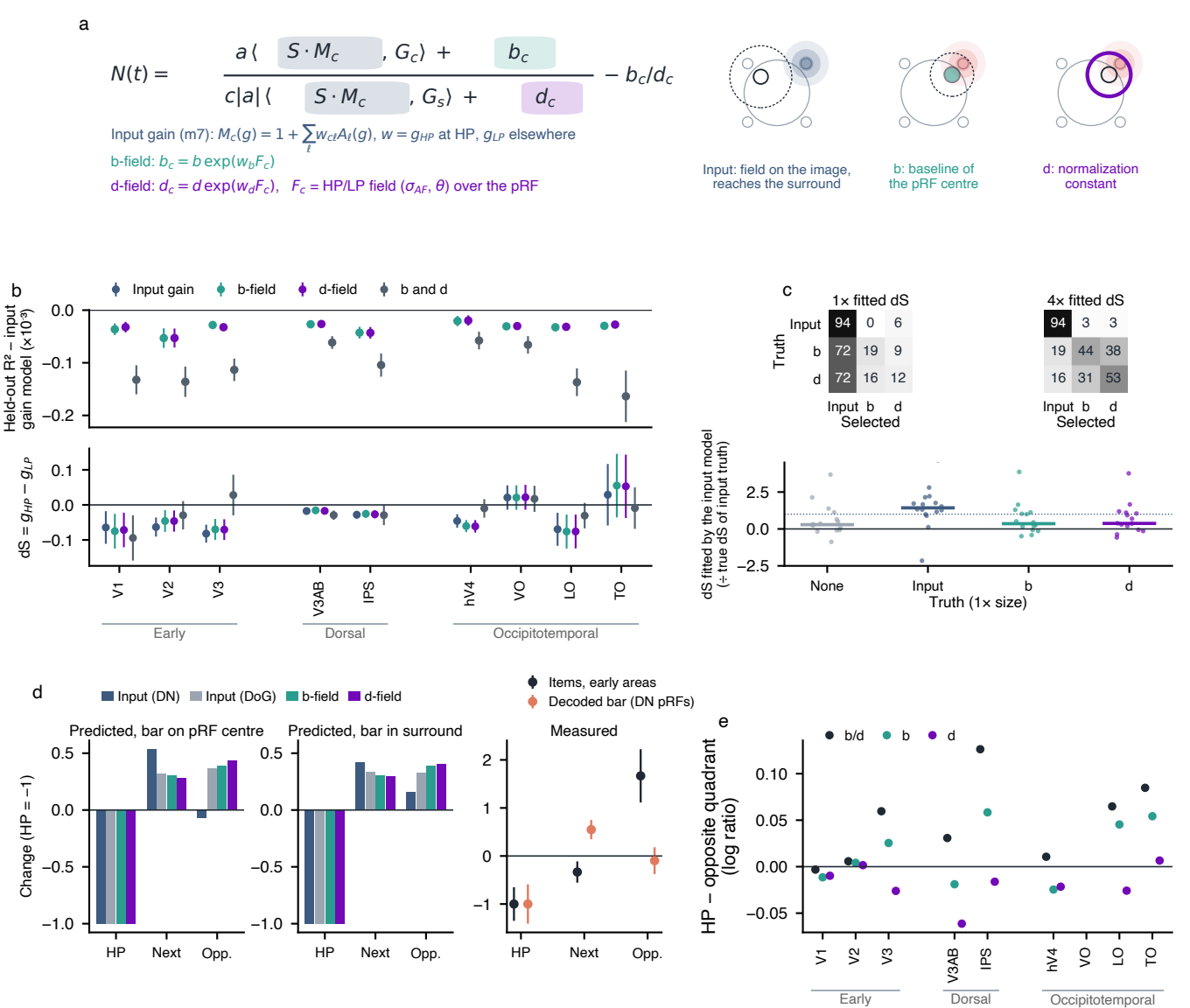

pRF

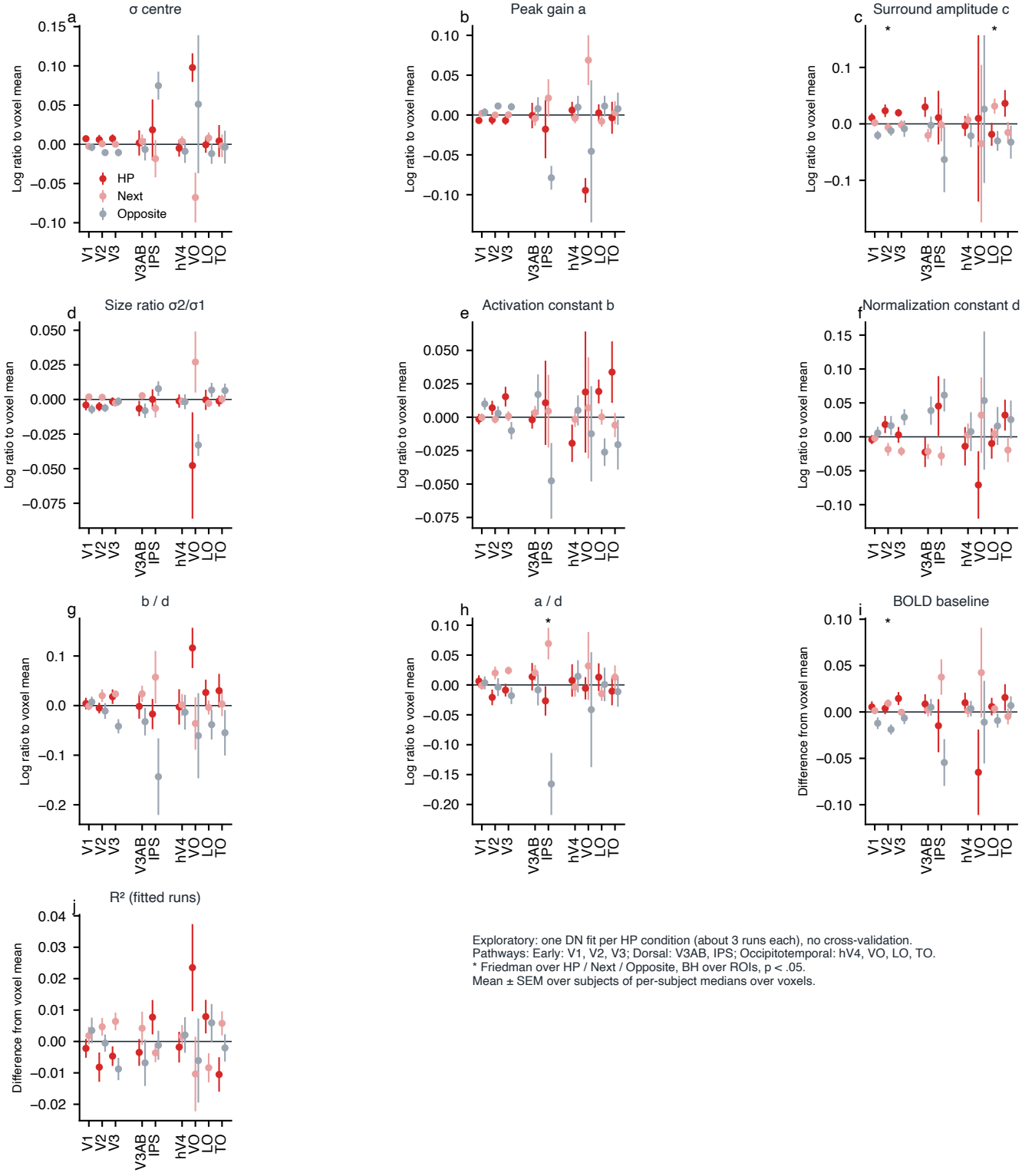

Bar crosses pRF
0
10 s


V1–V3: each location minus the mean of the four
Δ response vs mean of 4 locations (% signal)
.05
0
−.05
**
HP
Next to HP
Opposite
0
5
10
Time from crossing (s)


a
Gain at the target beyond V1, at the distractor in V2, V3AB, hV4
Gain
1.5
1.0
0.5
0.0
V1
V2
V3
V3AB
IPS
hV4
VO
LO
TO
Early
Dorsal
Occipitotemporal
b
Size of the gain fields
5.7°
LP
HP
FWHM, median 3.3°
Median per ROI
Sustained field, to scale
Sustained
Transient (target, distractor)
Item spacing, 5.7°
FWHM (°)
20
10
5
2
c
Sustained gain: less at HP
Whole block, at HP – at LP
Gain, HP – LP
0.05
0.00
−0.10
d
Transient gain, HP – LP: distractor and target
Each trial
As Fig. 1c, d
0.2
0.0
−0.2


a
Gain at the target in 7 of 9 areas, at the distractor in V3AB
Gain
1.5
1.0
0.5
0.0
V1
V2
V3
V3AB
IPS
hV4
VO
LO
TO
Early
Dorsal
Occipitotemporal
b
Size of the gain fields
5.7°
LP
HP
FWHM, median 3.1°
Median per ROI
Sustained field, to scale
Sustained
Transient (target, distractor)
Item spacing, 5.7°
FWHM (°)
20
10
5
2
c
Sustained gain: less at HP
Whole block, at HP – at LP
Gain, HP – LP
0.05
0.00
−0.10
d
Transient gain, HP – LP: distractor and target
Each trial
As Fig. 1c, d
Gain, HP – LP
0.2
0.0
−0.2


a
$N(t) = \frac{a\langle S \cdot M_c, G_c\rangle + b_c}{c|a|\langle S \cdot M_c, G_s\rangle + d_c} - b_c/d_c$
Input gain (m7): $M_c(g) = 1 + \sum_t w_{ct} A_t(g)$, $w = g_{HP}$ at HP, g_{LP} elsewhere
b-field: $b_c = b\exp(w_b F_c)$
d-field: $d_c = d\exp(w_d F_c)$, F_c = HP/LP field (σ_{AF}, θ) over the pRF
Input: field on the image, reaches the surround
b: baseline of the pRF centre
d: normalization constant
b
Input gain
b-field
d-field
b and d
Held-out R² – input gain model (×10³)
dS = $g_{HP} - g_{LP}$
V1
V2
V3
V3AB
IPS
hV4
VO
LO
TO
Early
Dorsal
Occipitotemporal
c
1× fitted dS
4× fitted dS
Truth
Selected
Input
b
d
dS fitted by the input model (÷ true dS of input truth)
None
Truth (1× size)
d
Input (DN)
Input (DoG)
b-field
d-field
Predicted, bar on pRF centre
Predicted, bar in surround
Change (HP = −1)
HP
Next
Opp.
Items, early areas
Decoded bar (DN pRFs)
Measured
e
b/d
HP – opposite quadrant (log ratio)



Exploratory: one DN fit per HP condition (about 3 runs each), no cross-validation.
Pathways: Early: V1, V2, V3; Dorsal: V3AB, IPS; Occipitotemporal: hV4, VO, LO, TO.
* Friedman over HP / Next / Opposite, BH over ROIs, $p < .05$.
Mean ± SEM over subjects of per-subject medians over voxels.